

AGML Lab: Final Project

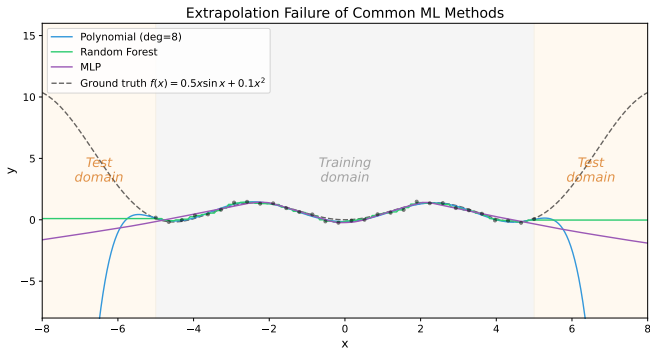
Equation Discovery



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One aim of machine learning is predicting data outside of the observed domain.



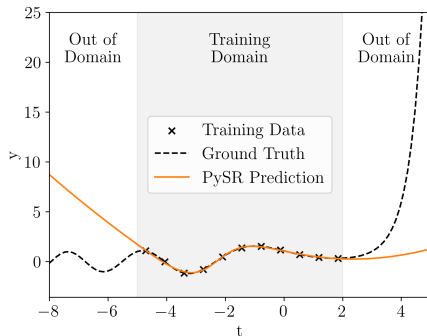
But most of our ML methods are remarkably bad at extrapolation!

Traditionally the natural sciences had much success describing nature with the help of formulas.

Once a system is found to follow a set formula, the formula serves as a very robust model for future behaviour.

Symbolic Regression aims to harness this, by looking for a formula that describes data well.

If the data does not follow a symbolic formula then symbolic regression can not be expected to extrapolate well.



Ground truth without symbolic description (numerical solution of airy equation) vs Symbolic Regressor

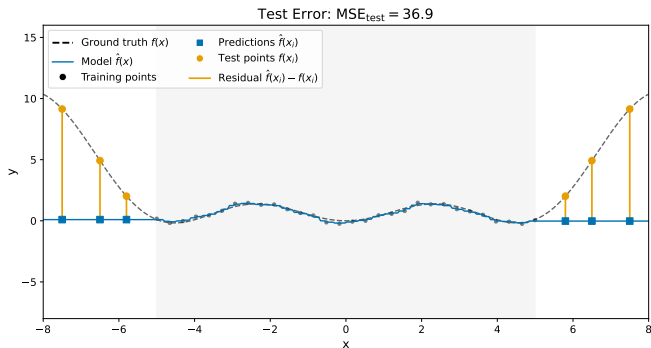
Luckily, we humans designed these formulas to make abstract models of the world.

Therefore a good portion of the world can be described by formulas.

1. Build an algorithm that extrapolates well on data generated by a symbolic formula.
 - ▶ whether your algorithm uses pre-training, fits a simple regression model, or performs search is up to you
 - ▶ i provide literature on moodle
2. Find your algorithm's strengths and weaknesses with your own experiments.
3. Write a report about it.
 - ▶ ≤ 7 pages (excluding references)
 - ▶ due 30th of August but feel free to finish early
 - ▶ send me your code
 - ▶ i provide template and evaluation criteria on moodle
4. Come to my office and tell me about your project for the oral evaluation.
 - ▶ you should be able to explain relevant parts of your project
 - ▶ i'll make an exam appointment with you when you hand in the report, preferrably within the next week

Your minimization goal should be how bad your predictions fail outside of the training domain, for example with a MSE:

$$\text{MSE}_{\text{test}} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} (\hat{f}(x_i) - f(x_i))^2$$



I will provide a small unlabeled dataset you can get started on, but you should generate most of your data yourself:

1. sample a function by rolling dice with operators
2. sample outputs from that function to generate train + test data

Feel free to find your own datasets where you expect symbolic relationships, and include them in your report! That would be cool (but it's not necessary).
